MIL-T-45071A(Ord)
11 July 1960

SUPERSEDING
MIL-T-45071 (Ord)
20 May 1958

MILITARY SPECIFICATION

TELESCOPES: M102 AND M102C (ARTICULATED); PACKAGING OF

1. SCOPE

1.1 This specification covers the packaging and packing of Telescopes M102 and M102C (Articulated) for each level of protection.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-E-760	Ethyl Alcohol (Ethanol)
NN-P-515	Plywood; Container Grade
UU-P-313	Paper, Lens Tissue
UU-T-106	Tape, Masking; Paper (Pressure-Sensitive)
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)
CCC-C-440	Cloth, Cotton, Cheesecloth, Bleached and Unbleached
JJJ-C-561	Cotton; Sterile and Nonsterile
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-C-843	Cushionging Material, Cellulosic
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

FSC 1240

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Military

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-B-121	Barrier Material, Greaseproofed, Flexible (Waterproofed)
MIL-B-130	Barrier Material, Paper, Noncorrosive
MIL-C-3133	Cellular Rubber Products, General Purpose (Except Ebonite)
MIL-C-7769	Cushioning Material, Bound
MIL-P-13988	Paper, Lens Tissue, Antitarnish, Wrapping

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and methods.- The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

3.1.1 Materials.- The following shall apply wherever packaging materials are referred to in this specification by name only:

REFERENCE	FULL IDENTIFICATION
Bound cushioning material	Type IV of Specification MIL-C-7769
Filament tape	Type III, at least 3/4 inch wide, of Specification PPP-T-97
Greaseproofed barrier material	Type I, grade A, class 2 of Specification MIL-B-121
Gummed tape	Class 2 of Specification UU-T-111
Noncorrosive barrier material	Type II of Specification MIL-B-130
Paperboard cushioning	Type I, class 1 of Specification PPP-P-291
Pressure-sensitive tape	Type I or II of Specification UU-T-106

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3.2 Size of containers.- The bag and exterior container shall be of a minimum size that will adequately contain the item.

3.3 Packaging.

3.3.1 Level A.

3.3.1.1 Disassembling.- The headrest assembly shall be removed from the telescope. The knob and washers, which secure the headrest assembly to the telescope, shall be placed back in their respective positions on the telescope.

3.3.1.2 Cleaning.

3.3.1.2.1 Optics.

3.3.1.2.1.1 The following materials and equipment shall be used in the cleaning of optics:

Materials and equipment	Specification
Alcohol, ethyl	Grade II of Specification O-E-760
Bottle, glass, 4 ounce	----------
Brush, camel's hair	----------
Cheesecloth	Specification CCC-C-440
Cloth, batiste	----------
Cotton, absorbent	Grade A, class I of Specification JJJ-C-561
Paper, lens tissue	Specification UU-P-313
Sticks, orange or equal	----------
Syringe, rubber	----------

3.3.1.2.1.2 Cleaning the exposed optics shall be accomplished in the following manner:

(a) Loose particles of dust shall be removed from the surfaces of the optics with a camel's hair brush.

(b) The surface to be cleaned shall be wiped with a circular motion using cotton, lens paper or freshly laundered cheesecloth which is saturated with ethyl alcohol. The surface shall be dried with clean cotton, lens paper or freshly laundered cheesecloth.

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(c) A swab shall be made by wrapping a piece of lens tissue paper around the end of an orange stick or equal. One or two drops of alcohol shall be added on the tip of the swab. The swabbing operation shall begin at the center of the polished surface. A light, downward pressure shall be exerted on the swab and with a circular motion gradually increase the radius of area being cleaned until the entire surface has been swabbed. This operation shall be repeated until no trace of dirt, lint or smears are perceptible.

(d) A rubber syringe shall be used as a bellows to remove any residual dust or lint that may remain on the cleaned surface.

(e) Immediately after cleaning, the metal cap shall be placed in position on the eyepiece assembly and the objective end of the telescope shall be covered with lens tissue paper conforming to Specification MIL-P-13988, which shall be secured in place with pressure-sensitive tape.

3.3.1.2.2 Other surfaces.- Cleaning Process C-1 of Specification MIL-P-116 shall be followed to clean all other surfaces.

3.3.1.3 Drying.- Immediately after cleaning, items shall be dried thoroughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the items.

3.3.1.4 Unit packaging.- Each telescope shall be packaged as specified herein, which is in conformance with Method IA-8 of Specification MIL-P-116.

3.3.1.4.1 Covering the tissue covered lens.- The tissue paper covering the objective end of the telescope shall be covered with cellulosic cushioning conforming to any type, class B or C (at least 1/4 inch thick) of Specification PPP-C-843. The cushioning material shall be secured in place with pressure-sensitive tape.

3.3.1.4.2 Mounting hardware.- The mounting hardware for securing the hanger assembly in the tank shall be wrapped with noncorrosive barrier material. The wrap shall be secured in place with pressure-sensitive tape.

3.3.1.4.3 Support and hanger assembly.- A piece of greaseproofed barrier material shall be secured to the support and hanger assembly with pressure-sensitive tape and a rubber pad shall be placed on the greaseproofed barrier material and secured in place with pressure-sensitive tape in such a manner that when the support and hanger assembly is rotated to the position shown in figure 1, the rubber covered portion of the assembly will be resting on the telescope. The

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assembly shall be secured to the telescope with filament tape in such a fashion that the assembly is completely immobilized. The piece of greaseproofed barrier material shall be approximately four inches square. The rubber pad shall be approximately four inches square and one-half inch thick and shall conform to grade RN13AF or RN14AF of Specification MIL-C-3133. The wrapped mounting hardware shall be secured to the hanger and support assembly with pressure-sensitive tape as shown in figure 1.

3.3.1.4.4 Headrest assembly.- The headrest assembly shall be wrapped with paperboard cushioning which shall be secured in place with gummed tape. The wrapped assembly shall then be secured to the telescope with filament tape as shown in figure 1 for Telescope M102.

3.3.1.4.5 Covering opening for instrument light.- The opening for the instrument light shall be covered with pressure-sensitive tape.

3.3.1.4.6 Wrapping telescope.- The telescope shall be wrapped with non-corrosive barrier material. The wrap shall be secured in place with pressure-sensitive tape. The wrapped telescope shall be overwrapped with paperboard cushioning which shall be secured in place with gummed tape.

3.3.1.4.7 Water-vaporproof bag.- The wrapped telescope shall be placed in a water-vaporproof bag conforming to type I, class e of Specification MIL-B-117. The bag shall be heat sealed in such a manner that it will pass the heat seal and quick leak tests of Specification MIL-P-116. Air shall be exhausted from the bag prior to final closure.

3.3.2 Level C.

3.3.2.1 Disassembling, cleaning and drying.- Disassembling, cleaning and drying shall be as specified in 3.3.1.1, 3.3.1.2 and 3.3.1.3.

3.3.2.2 Unit packaging.- Each telescope shall be packaged as specified in 3.3.1.4.1 to 3.3.1.4.6 inclusive, which is in conformance with Method III of Specification MIL-P-116.

3.4 Packing.

3.4.1 Level A.

3.4.1.1 Shipping container.- The shipping container, in which the unit packaged telescope is cushioned and immobilized (specified in 3.4.1.2) shall conform to one of the following specifications:

PPP-B-601 - Style A or B (oversea type) - type II or III, class 2 plywood
PPP-B-621 - Class 2 (oversea), style 2 or 4

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3.4.1.2 Cushioning.- The unit packaged telescope shall be packed in the shipping container in such a manner that there is at least three inches of bound cushioning material between the maximum projection on each side and top of the unit package and the interior surfaces of the container, and at least four inches of bound cushioning material between the maximum projection on the bottom of the unit package and the bottom surface of the container. Each end of the unit packaged telescope shall be cushioned in the following manner, which is shown in Figure 3:

(a) Bound cushioning material, approximately one inch thick, shall be placed at each end of the unit packaged telescope.

(b) A piece of plywood, approximately 3/8 inch thick, shall be placed next to each one inch piece of bound cushioning material. The plywood shall conform to any type, except type IV, class 2 of Specification NN-P-515.

(c) Bound cushioning material, approximately four inches thick, shall be placed between each piece of plywood and each end panel of the shipping container.

(d) The length and width of the one inch and four inch thick bound cushioning material and the pieces of plywood shall be approximately equal to the inside width and depth of the shipping container.

3.4.1.2.1 All voids shall be filled with bound cushioning material. The bound cushioning material shall be compressed in such a fashion that the unit packaged telescope fits snugly in the cushioning material when the shipping container is closed.

3.4.2 Level B.- One unit packaged telescope shall be cushioned and immobilized in a shipping container as specified in 3.4.1.2 and 3.4.1.2.1. The shipping container shall conform to one of the following specifications:

PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (Domestic), style 2 or 4

3.4.3 Level C.- One unit packaged telescope shall be cushioned and immobilized in a shipping container as specified in 3.4.1.2 and 3.4.1.2.1. The shipping container shall conform to one of the following specifications:

PPP-B-585 - Any style, except style 1, class 1, construct for type 3 load
PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic), any style

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3.4.4 Strapping.- Strapping shall be applied on the shipping container as specified in the applicable container specification.

3.5 Marking.

3.5.1 Interior packages.- Marking of interior packages shall be in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A unit package and omitted on the Level C unit package.

3.5.2 Shipping containers.- Each shipping container shall be marked in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A or B shipping container and omitted on the Level C container. Fragile labels shall be affixed to the container in accordance with Standard MIL-STD-129.

3.5.3 Stock number.- The stock number to be added in the identification data shall be obtained from the contracting officer.

3.6 Preproduction pack.- Prior to quantity packaging production, a preproduction pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification. A rough handling test is required.

3.7 Workmanship.- All operations involved in accomplishing corrosion prevention, preservation, packaging, and packing by the procedures specified herein shall be in accordance with the highest grades of practice associated with this type of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Contractor's quality assurance system . The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure required by this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

4.3 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have a detrimental affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

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4.4 Preproduction pack.- One preproduction pack shall be inspected by the Government for conformance to the requirements outlined herein. The preproduction pack shall be representative of the manufacturer's normal production.

4.4.1 Disposition of rejected preproduction pack.- Should the preproduction pack fail to pass the test for Government acceptance, the contractor shall cease those operations which are causing rejectable material and make the necessary corrections. When corrections have been made, a new preproduction pack shall be tested by the contractor under surveillance of the Government. Operations causing rejections shall not be resumed until a preproduction model has successfully passed all inspections specified.

4.5 Inspection provisions.

4.5.1 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.5.2 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.3 Acceptance tests.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

4.5.3.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL:

None defined

MAJOR: AQL 2.5%		Req't Par.	Test Method
101.	Incorrect disassembling of telescopes (Levels A, C)	3.3.1.1 3.3.2.1	Visual
102.	Improper cleaning (Levels A, C)	3.3.1.2 3.3.2.1	Visual
103.	Improper drying (Levels A, C)	3.3.1.3 3.3.2.1	Visual
104.	Incorrect unit packaging method IA-8 and III (Levels A, C)	3.3.1.4 3.3.2.2	MIL-P-116
105.	Lens not covered with cellulosic cushioning, improper type, not secured, incorrect thickness (Levels A, C)	3.3.1.4.1 3.3.2.2	PPP-C-843 Visual
106.	Mounting hardware wrapping omitted, not secured (Levels A, C)	3.3.1.4.2 3.3.2.2	Visual
107.	Support and hanger assembly improperly packed, not secured (Levels A, C)	3.3.1.4.3 3.3.2.2	Figure 1 Visual
108.	Headrest assembly improperly wrapped (Levels A, C)	3.3.1.4.4 3.3.2.2	Figure 1 Visual
109.	Cover opening for instrument light not taped (Levels A, C)	3.3.1.4.5 3.3.2.2	Visual
110.	Telescope improperly wrapped, (Levels A, C)	3.3.1.4.6 3.3.2.2	Visual

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TABLE I - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 2.5%		Req't Par.	Test Method
111.	Water-vaporproof bag damaged, not according to specifications, excess air not exhausted (Level A)	3.3.1.4.7	MIL-B-117 MIL-B-116
112.	Shipping containers not conforming to specifications, unit packages not immobilized (Levels A, B, C)	3.4.1 3.4.2 3.4.3	PPP-B-585 PPP-B-601 PPP-B-621 Visual
113.	Strapping incorrectly applied (Levels A, B, C)	3.4.4	Visual
114.	Markings illegible, incorrect, missing, not conforming to specifications (Levels A, B, C)	3.5	Visual MIL-STD-129
115.	Workmanship (Levels A, B, C)	3.7	Visual

4.6 Test methods and procedures.

4.6.1 Rough handling test.- The preproduction pack only (specified in 3.6) shall be subjected to the rough handling test in accordance with Specification MIL-P-116.

4.6.2 Sampling for test.- Samples for the tests listed in Table II shall be selected from lots that have met the minimum inspection requirements listed in Table I.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL:

None defined.

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TABLE II - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 4.0%		Req't Par.	Test Method
1.	Heat seal	3.3.1.4.7	MIL-P-116 IA-8
2.	Quick leak	3.3.1.4.7	MIL-P-116 IA-8
3.	Determination of cleanliness	3.3.1.2 3.3.2.1	IA-8, III MIL-P-116

5. PREPARATION FOR DELIVERY

5.1 There are no applicable requirements.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of specification.
(b) Selection of applicable levels of packaging and packing.

6.2 Size and weight of shipping container containing a packaged telescope.- The following inside dimensions, overall cubage and gross weight are approximate for one unit packaged telescope packed in a nailed wood box for Level A:

Inside dimensions - 69. inches long by 16. inches wide by 17. inches deep
Cubage - 15.8 cubic feet
Gross weight - 200. pounds

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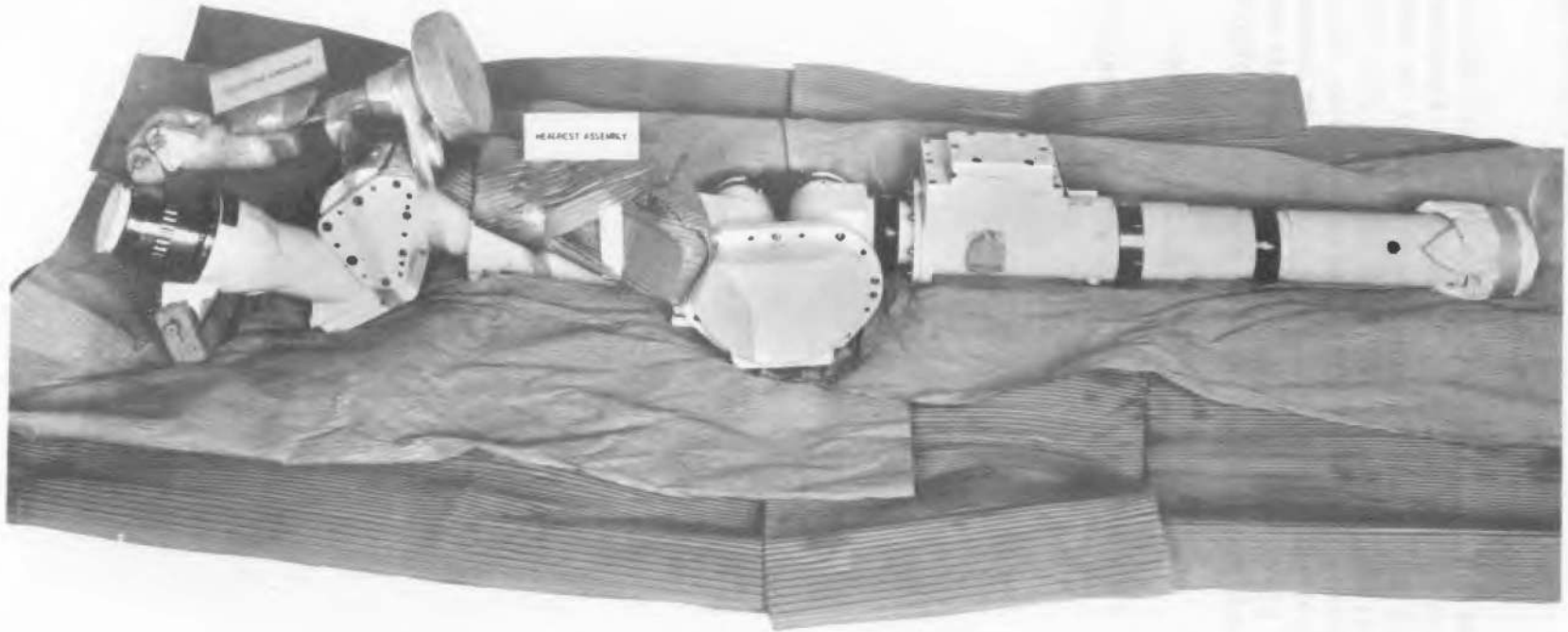
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Preparing Activity:

Army - Ordnance Corps

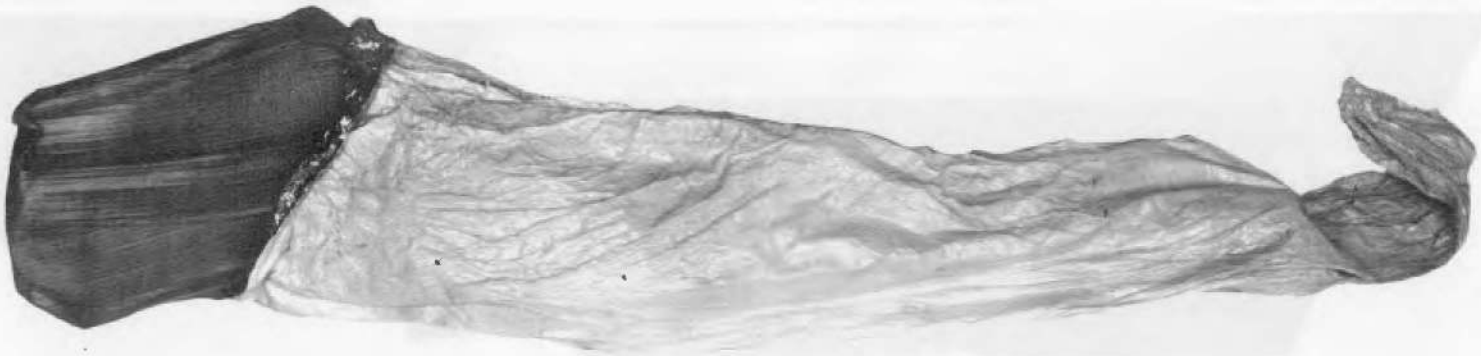
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TELESCOPE M102 - Showing wrapped hardware, wrapped headrest and the support and hanger assembly (wood mock up) secured in place prior to wrapping the telescope.

FIGURE 1

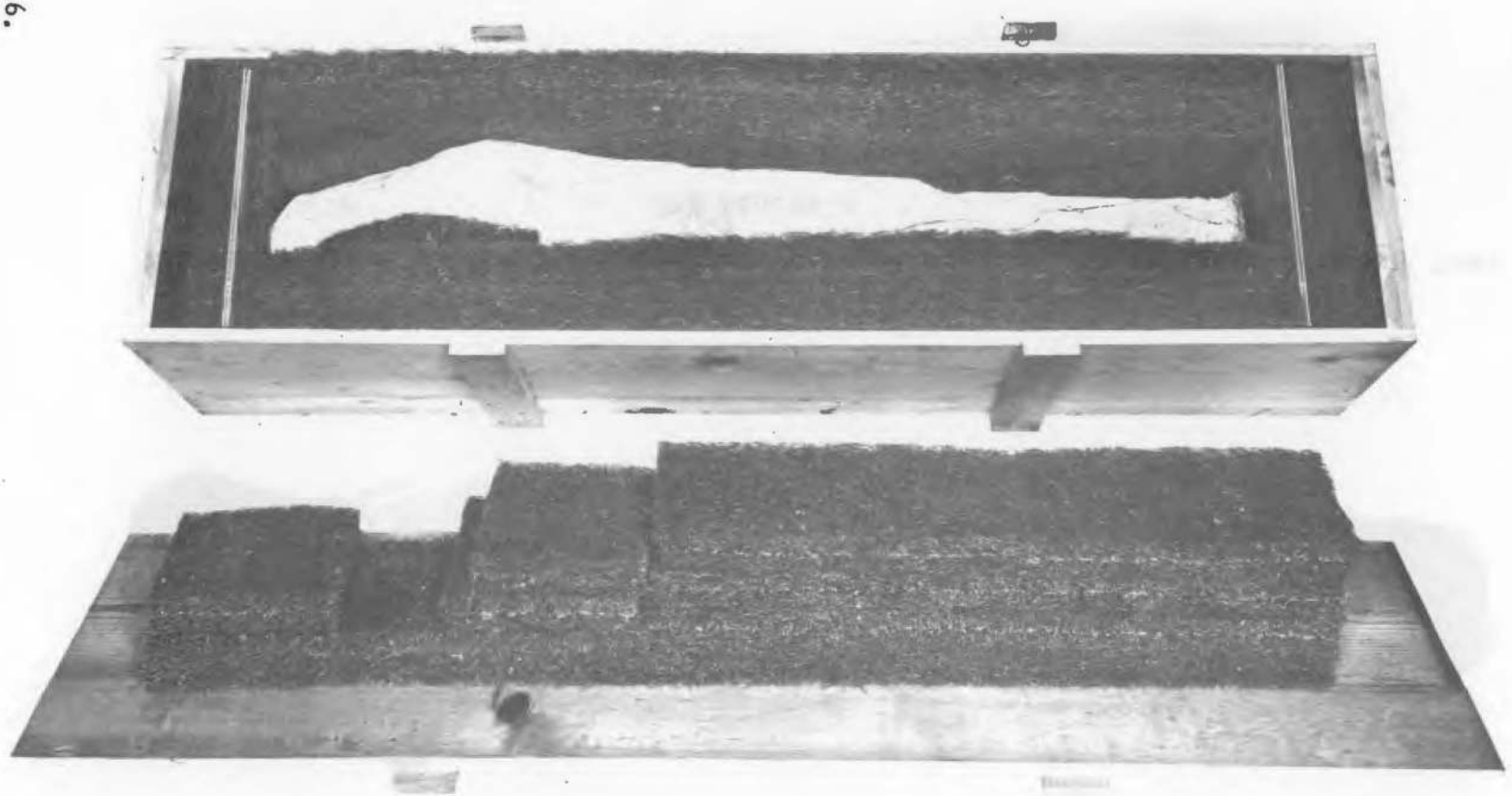
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TELESCOPE M102 - Showing wrapped telescope being placed in water-vaporproof bag.

FIGURE 2

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TELESCOPE M102 - Showing bagged telescope being immobilized in bound cushioning material in shipping container

FIGURE 3

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